DESCRIPTION OF COURSEWORK

Course Code	EEE304
Course Name	High Voltage Engineering
Lecturer	Ir Dr Mohd Faris bin Abdullah
Academic Session	09/2023
Assessment Title	Impulse Voltage Generation and Electrical Breakdown Under Impulse Voltage

A. Introduction/ Situation/ Background Information

Introduction

Figure 1 shows a typical impulse voltage waveform, where T is the time from 0.3 to 0.9 of the peak V, T_1 =1.67T is the 'front time' and T_2 = time from O_1 to 0.5V is the 'tail time.' The international standard lightning impulse voltage is 1.2/50μs, where 1.2μs is the front time T_1 and 50μs is the tail time T_2

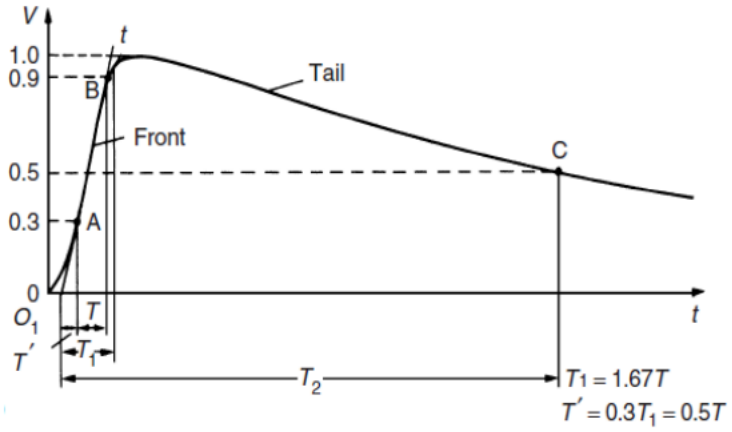


Figure 1

Objectives

1. To construct and simulate impulse voltage generation circuit
2. To measure breakdown voltage in air gap under impulse voltage
3. To investigate the effect of electrode shapes on impulse breakdown voltage

Location

UM High Voltage Laboratory, Department of Electrical Engineering, Faculty of Engineering, Universiti Malaya, 50603 Kuala Lumpur, Malaysia

Equipment

High voltage generation kit, air gap, electrodes

Precautions

Follow the demonstrator's instructions throughout the experiment

References

Refer to materials related to High Voltage Engineering

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Analyze the methods of generations and measurements of high voltage AC, DC and impulse generator.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a

serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group of **four (4)** student laboratory to be completed in 2 weeks.

One (1) report in softcopy (PDF) should be submitted and follow the following format:

1. Methodology
2. Results and Discussion

Due Date: 30th November 2023 (Week 11)

Late submission will not be accepted.

Software Tool:

Multisim, Pspice, Psim, Matlab/Simulink, EMTP-ATP and PSCAD

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Construct and simulate equivalent circuit of impulse voltage generation that you have connected	40
2.	Plot the measured breakdown voltage (kV) and calculated breakdown voltage (kV) versus air gap distance (mm) in one graph for all electrode shapes. Compare and discuss the measured and calculated breakdown voltage	40
3.	Discuss and evaluate the effect of electrode shapes on impulse breakdown voltage	20
	TOTAL	**100**

F. Task(s)

Impulse Voltage Generation

1. Construct and simulate equivalent circuit of impulse voltage generation that you have connected by using simulation software such as Multisim, Pspice, Psim, Matlab/Simulink, EMTP-ATP and PSCAD

Experiment Procedures

1. Refer to the manuals of the high voltage kit and connect the relevant components so that impulse high voltage can be generated
2. Connect the air gap to the impulse high voltage output
3. Fix the air gap with a sharp point electrode on the top part and a large plane electrode on the bottom part
4. Referring to Table 1, measure the breakdown voltage of each gap distance

Air Gap Distance (mm)	Calculated Breakdown Voltage (kV)	Measured Breakdown Voltage (kV)		
		Sharp Point - Plane Electrode	Sphere Point - Plane Electrode	Cylindrical Point - Plane Electrode
2				
4				
6				
8				
10				

Table 1

5. Observe and note down the behaviour or pattern of the arc across the air gap under impulse voltage
6. Reduce the applied voltage amplitude and switch off the power supply
7. Change the upper part of the gap with a sphere electrode while maintaining the bottom part with a plane electrode. Repeat step 4
8. Change the upper part of the gap with a cylindrical electrode while maintaining the bottom part with a plane electrode. Repeat step 4
9. Calculate the breakdown voltage of each gap in Table 1 using a theoretical formula given by

$$V = \frac{a(pd)}{\ln(pd) + b} \quad \text{[in kV]}$$

where p is the air pressure in Atm, d is the air gap distance in meters, a = 43,600 and b = 12.8 for air

Results and Discussions

1. Construct and simulate the equivalent circuit of impulse voltage generation that you have connected. Discuss and analyze the impulse voltage simulated. Display the simulation snapshot results in your report
2. Plot the measured breakdown voltage (kV) and calculated breakdown voltage (kV) versus air gap distance (mm) in one graph for all electrode shapes. Compare and discuss the measured and calculated breakdown voltage
3. Discuss and evaluate the effect of electrode shapes on impulse breakdown voltage

APPENDIX 1

MARKING RUBRICS

Component Title	**Impulse Voltage Generation and Electrical Breakdown Under Impulse Voltage**				**Percentage (%)**		
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Construct and simulate the equivalent circuit of impulse voltage generation that you have connected	100% equivalent circuit of impulse voltage generation is constructed and simulated with comprehensively and correctly	80% equivalent circuit of impulse voltage generation is constructed and simulated with comprehensively and correctly	60% equivalent circuit of impulse voltage generation is constructed and simulated with comprehensively and correctly	40% equivalent circuit of impulse voltage generation is constructed and simulated with comprehensively and correctly	Less than 40% equivalent circuit of impulse voltage generation is constructed and simulated with comprehensively and correctly	8	
Plot the measured breakdown voltage (kV) and calculated breakdown voltage (kV) versus air gap distance (mm) in one graph for all electrode shapes. Compare and discuss the measured and calculated breakdown voltage	100% measured and calculated breakdown voltage are plotted and correctly compared and discussed	80% measured and calculated breakdown voltage are plotted and correctly compared and discussed	60% measured and calculated breakdown voltage are plotted and correctly compared and discussed	40% measured and calculated breakdown voltage are plotted and correctly compared and discussed	less than 40% measured and calculated breakdown voltage are plotted and correctly compared and discussed	8	
Discuss and evaluate the effect of electrode shapes on impulse breakdown voltage	100% effect of electrode shapes on impulse breakdown voltage are discussed with comprehensively and correctly evaluated	80% effect of electrode shapes on impulse breakdown voltage are discussed with comprehensively and correctly evaluated	60% effect of electrode shapes on impulse breakdown voltage are discussed with comprehensively and correctly evaluated	40% effect of electrode shapes on impulse breakdown voltage are discussed with comprehensively and correctly evaluated	less than 40% effect of electrode shapes on impulse breakdown voltage are discussed with comprehensively and correctly evaluated	4	
						TOTAL	

Note to students: Please include the marking rubric when submitting your coursework.